

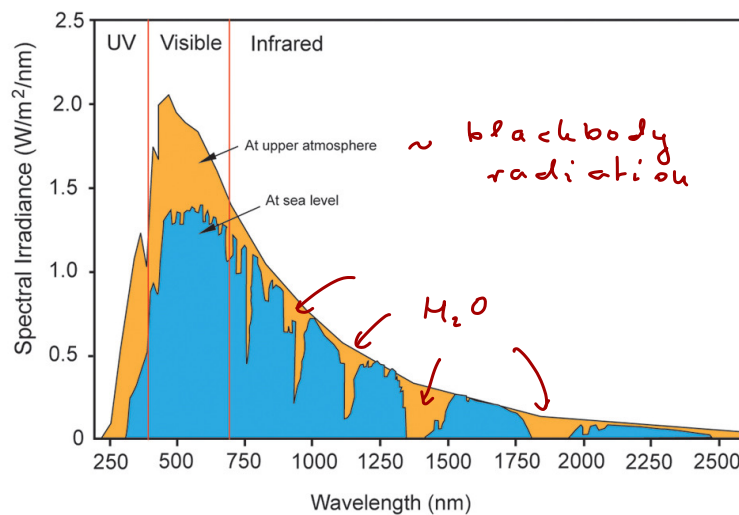
Fourier Analysis

Lectures 1-4

Fourier series / expansion

⇒ spectral analysis

Energy Spectrum of Sunlight



- light / radiation

- x-ray
- infrared
- ...

- electrical noise

- mechanical noise (sound)

⇒ approximation scheme

periodic functions →

Fourier series

aperiodic functions →

Fourier transform

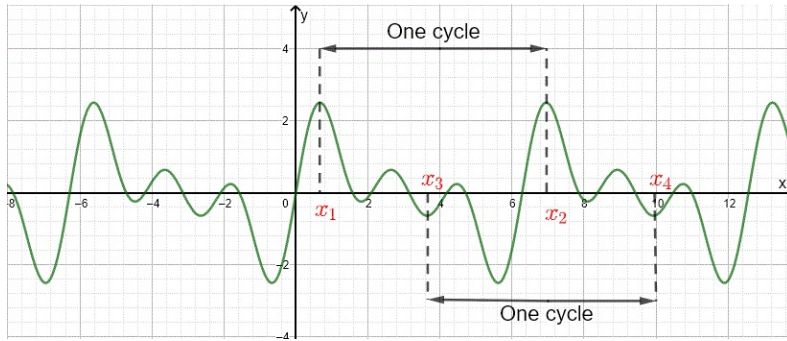
- linear differential equations

- partial " " "

Fourier Series

~ approximating by sines / cosines

periodic function: ~ a.c. current



Def. $f(x)$ is p -periodic if

$$f(x+p) = f(x)$$

p -periodic $\Rightarrow n \cdot p$ -periodic

$$f(x + n \cdot p) = f(x + (n-1) \cdot p) = \dots = f(x)$$

not periodic

$$x, x^2, \dots$$

periodic

$$\sin(x), \cos(x), \dots$$

$$p = 2\pi$$

$$\{x\} = x - [x]$$

fractional part

$$p = 1$$

$$\{x\}^2 = (x - [x])^2$$

trigonometric series

function series :

$$a_0 + a_1 \cos(x) + b_1 \sin(x) + a_2 \cos(2x) + b_2 \sin(2x) + \dots$$

$$= a_0 + \sum_{k=1}^{\infty} (a_k \cos(k \cdot x) + b_k \sin(k \cdot x))$$

Fourier coefficients

$$\bullet f_n(x) \equiv a_0 + \sum_{k=1}^n (a_k \cos(k \cdot x) + b_k \sin(k \cdot x))$$

2π -periodic functions

Claim: virtually any 2π -periodic function of interest can be represented as

$$(*) \quad f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(k \cdot x) + b_k \sin(k \cdot x))$$

Euler formulas

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} dx f(x) \cos(nx)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)$$

example:

$$f(x) = \begin{cases} 1 & 0 < x < \pi \\ -1 & -\pi < x < 0 \end{cases}$$

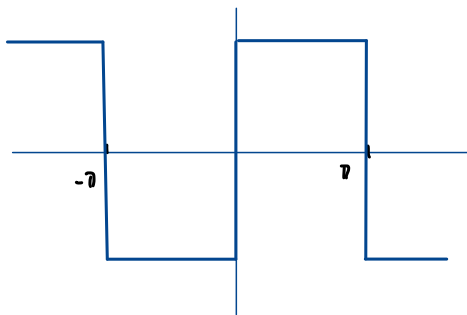
$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} dx f(x) = 0$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} dx f(x) \cos(nx) = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} dx \sin(nx) \cdot \text{sgn}(x) = \frac{2}{\pi} \int_0^{\pi} \sin(nx) dx$$

$$= \frac{2}{\pi \cdot n} \left[-\cos(nx) \right]_0^{\pi} = \frac{2}{\pi \cdot n} (1 - \cos(n\pi)) =$$

$$= \begin{cases} 0 & n \text{ even} \\ \frac{4}{\pi \cdot n} & n \text{ odd} \end{cases}$$



$$= \frac{4}{\pi} \sin(x) + \frac{4}{3\pi} \sin(3x) + \dots$$

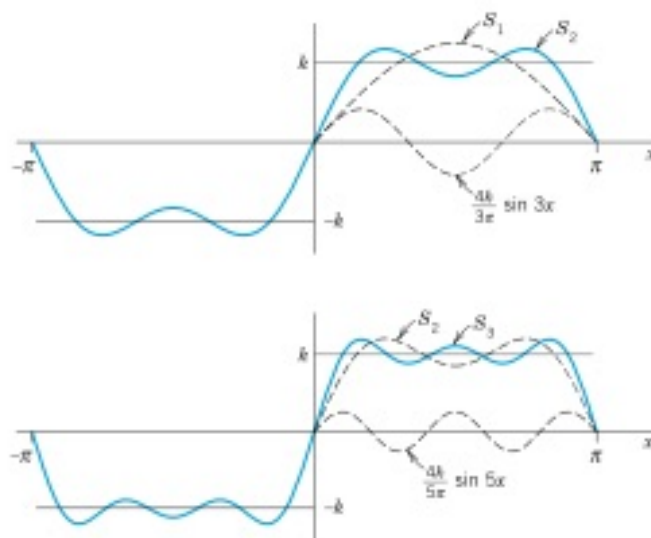


Fig. 261. First three partial sums of the corresponding Fourier series

Proof:

use integral identities

$$n, m \in \mathbb{Z} \setminus \{0\}$$

$$\int_{-\pi}^{\pi} dx \cos(n \cdot x) \cos(m \cdot x) = \pi \cdot \delta_{n,m}$$

$$\underbrace{\hspace{10em}}_{C_{n,m}}$$

Kronecker delta
 $= 1$ if $n = m$
 0 otherwise

use addition rule

$$\cos(\alpha) \cos(\beta) = \frac{1}{2} (\cos(\alpha + \beta) + \cos(\alpha - \beta))$$

$$C_{nm} = \frac{1}{2} \int_{-\pi}^{\pi} \left\{ \underbrace{\cos((n+m) \cdot x)}_{\substack{1 \text{ if } n+m=0 \\ \text{only possible} \\ \text{if } n=m=0}} + \underbrace{\cos((n-m) \cdot x)}_{1 \text{ if } n=m} \right\}$$

$$C_{00} = 2\pi \quad ; \quad C_{0n} = 0 \quad ; \quad C_{n,m} = \pi \cdot \delta_{n,m} \quad n, m \geq 1$$

$$\int_{-\pi}^{\pi} dx \underbrace{\sin(nx)}_{\text{odd}} \cdot \underbrace{\cos(mx)}_{\text{even}} = 0$$

$$\int_{-\pi}^{\pi} dx \sin(nx) \sin(mx) = \pi \cdot \delta_{n,m}$$

Now assume (*)

$$\Rightarrow \bullet \int_{-\pi}^{\pi} dx f(x) =$$

$$\int_{-\pi}^{\pi} dx \left(a_0 + a_1 \underbrace{\cos(x)}_{\neq} + b_1 \underbrace{\sin(x)}_{\neq} + a_2 \underbrace{\cos(2x)}_{\neq} + b_2 \underbrace{\sin(2x)}_{\neq} + \dots \right)$$

$$= a_0 \cdot 2\pi$$

$$\Rightarrow a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \quad \checkmark$$

$$\bullet \int_{-\pi}^{\pi} dx f(x) \cdot \sin(x) =$$

$$= \int_{-\pi}^{\pi} dx \left(a_0 + a_1 \underbrace{\cos(x)}_{\neq} + b_1 \sin(x) + a_2 \underbrace{\cos(2x)}_{\neq} + \dots \right) \cdot \sin(x)$$

$$= b_1 \int_{-\pi}^{\pi} \sin^2(x) dx = b_1 \cdot \pi \Rightarrow b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} dx f(x) \cdot \sin(x)$$

$$\bullet \int_{-\pi}^{\pi} dx f(x) \cdot \cos(x) = a_1 \cdot \pi \Rightarrow a_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} dx f(x) \cdot \cos(x)$$

• • •

□

Thm:

$f(x)$ piecewise continuous, 2π -periodic,
 f' finite at all other points

$\Rightarrow$ Fourier series converges to $f(x)$ except
for discontinuity points.

• general proof is difficult

• for f , f' and f'' continuous $\rightarrow$
uniform convergence is easy to prove

Extension to $f(x+2L) = f(x)$

trivial: $\cos(n \cdot x) \rightarrow \cos(x \cdot n \cdot \frac{\pi}{L})$

$$\Rightarrow f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(n \cdot \frac{\pi}{L} x\right) + b_n \sin\left(\frac{n \cdot \pi}{L} \cdot x\right) \right)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L dx f(x)$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n \cdot \pi}{L} \cdot x\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \sin\left(\frac{n \cdot \pi}{L} \cdot x\right) dx$$

- Even and odd functions

$$f(x) \text{ even} \Rightarrow b_n \equiv 0 \quad ! \quad (\text{only cos})$$

$$f(x) \text{ odd} \Rightarrow a_n \equiv 0 \quad (\text{only sin})$$

- Fourier coefficients of $f_1 + f_2$ (both of same period) are SUMS of coefficients

Fourier coeffs of c.f are c. Fourier coeffs of f

Challenges :

- Statement : Fourier coefficients give best trigonometric approx.

consider

$f(x)$ some 2π -periodic function

approximate

$$F(x) \equiv A_0 + \sum_{n=1}^N (A_n \cos(nx) + B_n \sin(nx))$$

deviation

$$\|F - f\|^2 \equiv \int_{-\pi}^{\pi} dx (f(x) - F(x))^2$$

minimal if $A_n \rightarrow a_n, B_n \rightarrow b_n$

- Parseval

$$\int_{-1}^1 dx f^2(x) = \pi \left(2 \cdot a_0^2 + a_1^2 + b_1^2 + \dots \right)$$

- 'Scalar product'

$$\int_{-1}^1 dx f(x) \cdot g(x) = \pi \left(2 a_0 \tilde{a}_0 + \sum_{n \geq 1} (a_n \tilde{a}_n + b_n \tilde{b}_n) \right)$$

$\downarrow$
 a_n, b_n

$\downarrow$
 $\tilde{a}_n, \tilde{b}_n$